

Day -5 and 6 TASK REPORT

Date: 05-06-2026

Activities Completed

- Researched LLMs, RAG, and AI Agents. Here's what I learned and what I want to know.
- Researched LLMs to understand what they are, how they work, and their role in modern AI applications.
- Researched RAG (Retrieval-Augmented Generation) and learned where it is used, how it retrieves information from external knowledge sources, and how it improves the accuracy of AI-generated responses.
- Researched AI Agents and learned where they are used, their architecture, and how they can perform tasks autonomously with reasoning and decision-making capabilities.
- Compared LLMs, RAG, and AI Agents to understand their differences, advantages, limitations, and suitable use cases.
- Learned about how to approach learning these technologies and how they can be incorporated into future mini-projects.
- Learned the basics of each technology: LLMs, RAG, and AI Agents.
- Studied the limitations, best architecture practices, step-by-step learning roadmap, and core features of LLMs, RAG systems, and AI Agents.

Practical Applications and Use Cases Studied

- Explored how LLMs are used in AI chatbots, virtual assistants, content generation, coding assistants, and customer support systems.
- Learned how RAG systems are used in document-based question answering systems, enterprise knowledge bases, research assistants, and applications that require accurate and updated information.
- Studied how AI Agents are used for workflow automation, task execution, business process automation, personal assistants, and systems that interact with external tools and APIs.

- Understood which technology is best suited for different scenarios:
 - LLMs for language understanding and text generation.
 - RAG for information retrieval and knowledge-based responses.
 - AI Agents for autonomous task execution and decision making.
- Identified potential applications of these technologies in future AI-powered projects and software solutions.

Outcome of the Day

- Gained a clear understanding of LLMs, RAG systems, and AI Agents.
- Learned their real-world applications and project use cases.
- Understood their differences, limitations, and best practices.
- Developed foundational knowledge and a learning roadmap for further study in AI technologies.
- Learned where and when to use each technology based on project requirements.
- Improved understanding of modern AI architectures and industry implementations.

Reference Resources

- ChatGPT
- Gemini